

AOA2

Number Sense and Functions

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A sixth-grade teacher preparing a warm-up for a class of 28 students writes $18 - 6 \div 3 \times 2 + 4$ on the board. What is the value?

- A. 21
- B. 18
- C. 12
- D. 10

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

Where should one pair of parentheses go so that $18 - 6 \div 3 \times 2 + 4$ equals 12?

- A. $18 - 6 \div (3 \times 2) + 4$
- B. $18 - (6 \div 3 \times 2 + 4)$
- C. $(18 - 6 \div 3) \times 2 + 4$
- D. $(18 - 6) \div 3 \times 2 + 4$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

In a mountain town that sits at 9,000 feet and draws skiers every winter, the morning temperature was -7°F . By afternoon it was 5°F . How many degrees did the temperature rise?

- A. -12 degrees, because subtracting always gives a smaller number.
- B. 2 degrees
- C. -2 degrees
- D. 12 degrees

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A baker who opened her shop in 2019 has $\frac{3}{4}$ cup of glaze and uses $\frac{1}{8}$ cup on each tray of pastries. How many trays can she glaze?

- A. $\frac{1}{6}$
- B. 6
- C. $\frac{3}{32}$
- D. 4

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

A student computes $\frac{2}{3} \div 4$ and writes $\frac{8}{3}$. What is the error and the correct answer?

- A. The student added instead of multiplying; the answer is $\frac{6}{3}$.
- B. The student multiplied by 4 instead of by its reciprocal; the answer is $\frac{1}{6}$.
- C. No error; dividing by a whole number always makes the result larger.
- D. The student should have flipped $\frac{2}{3}$; the answer is $\frac{3}{8}$.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A school store run by the eighth-grade student council must add sales tax at a rate of 0.045 on each sale. How should the rate appear on the price sign as a percent?

- A. 0.45%
- B. 0.045%
- C. 45%
- D. 4.5%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A town library that moved into a new building this spring had 800 visitors in March and 1,000 visitors in April. What was the percent increase?

- A. 25%
- B. 80%
- C. 20%
- D. 200%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

After a 25% increase, a price is \$60. What was the original price?

- A. \$48
- B. \$45
- C. \$35
- D. \$75

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

In Room 14, students build a tile pattern: stage 1 uses 5 tiles, stage 2 uses 8, and stage 3 uses 11. If the pattern continues, how many tiles are in stage 20?

- A. 62
- B. 60
- C. 44
- D. 100

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A math coach observing a seventh-grade class that uses tablets reviews four sets of ordered pairs a student wrote. Which set is a function?

- A. $\{(2, 3), (2, 5), (4, 1)\}$
- B. $\{(0, 0), (1, 1), (1, -1)\}$
- C. $\{(1, 4), (2, 4), (3, 4)\}$
- D. None of them, since a function can never repeat an output.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

For $f(x) = 2x - 3$, $f(a) = 11$. What is a ?

- A. 19
- B. 4
- C. 7
- D. 11

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

During a lesson recorded for the teacher’s portfolio, a student in a class of 24 says 0.25 is greater than 0.4 because 25 is greater than 4. What is the best response?

- A.** They are equal, since both are less than one.
- B.** The student is right; decimals with more digits are always larger.
- C.** 0.4 is greater: written as 0.40 it is 40 hundredths, more than 25 hundredths.
- D.** 0.4 is greater because shorter decimals are bigger.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A sixth-grade teacher ~~preparing a warm-up for a class of 28 students~~ writes $18 - 6 \div 3 \times 2 + 4$ on the board. What is the value?

Target word: value of the expression **Decides:** Division and multiplication left to right before the subtraction and addition.

A	Confusion set	Multiplies 3×2 before dividing; $\times$ and $\div$ are done left to right.
B	Correct	$6 \div 3 = 2$, $\times 2 = 4$; $18 - 4 + 4 = 18$.
C	Dropped element	Works strictly left to right, dropping operation precedence.
D	Inverted operation	Adds $4 + 4$ before subtracting; $+$ and $-$ are done left to right.

2. Answer D Concept 1 • transfer

Where should one pair of parentheses go so that $18 - 6 \div 3 \times 2 + 4$ equals 12?

Target word: place grouping to hit a target value **Decides:** Grouping $18 - 6$ first gives $12 \div 3 \times 2 + 4$.

A	Confusion set	Gives $18 - 1 + 4 = 21$.
B	Inverted operation	Subtracts the whole group: $18 - 8 = 10$.
C	Adjacent category	Gives $16 \times 2 + 4 = 36$.
D	Correct	$12 \div 3 = 4$, $\times 2 = 8$, $+ 4 = 12$.

3. Answer D Concept 2

In a mountain town ~~that sits at 9,000 feet and draws skiers every winter~~, the morning temperature was -7°F . By afternoon it was 5°F . How many degrees did the temperature rise?

Target word: size of the rise **Decides:** $5 - (-7) = 5 + 7$.

A	Absolute language	Subtracting a negative increases the value.
B	Dropped element	Drops the negative sign on -7 .
C	Inverted operation	Adds $5 + (-7)$ instead of subtracting.
D	Correct	Change = final – initial = $5 - (-7) = 12$.

4. Answer B Concept 3

A baker ~~who opened her shop in 2019~~ has $\frac{3}{4}$ cup of glaze and uses $\frac{1}{8}$ cup on each tray of pastries. How many trays can she glaze?

Target word: how many $\frac{1}{8}$ s in $\frac{3}{4}$ **Decides:** $\frac{3}{4} \div \frac{1}{8} = \frac{3}{4} \times \frac{8}{1}$.

A	Confusion set	Flips the dividend instead of the divisor.
B	Correct	$\frac{3}{4} \times 8 = \frac{24}{4} = 6$.
C	Inverted operation	Multiplies $\frac{3}{4} \times \frac{1}{8}$ instead of dividing.
D	Adjacent category	Subtracts denominators, a whole-number habit that does not apply to fractions.

5. Answer B Concept 3 • transfer

A student computes $2/3 \div 4$ and writes $8/3$. What is the error and the correct answer?

Target word: diagnose the error **Decides:** Divided by 4 should be multiplied by $1/4$.

A	Confusion set	Mislabeled the error and still gives a wrong value.
B	Correct	$2/3 \times 1/4 = 2/12 = 1/6$.
C	Absolute language	Dividing by a number greater than 1 makes the result smaller.
D	Inverted operation	Flips the dividend instead of the divisor.

6. Answer D Concept 4

A school store ~~run by the eighth grade student council~~ must add sales tax at a rate of 0.045 on each sale. How should the rate appear on the price sign as a percent?

Target word: decimal to percent **Decides:** Multiply by 100: move the point two places right.

A	Inverted operation	Moves the point left instead of right.
B	Dropped element	Adds a % sign without converting.
C	Confusion set	Reads 0.045 as 0.45.
D	Correct	$0.045 \times 100 = 4.5$.

7. Answer A Concept 5

A town library ~~that moved into a new building this spring~~ had 800 visitors in March and 1,000 visitors in April. What was the percent increase?

Target word: percent increase **Decides:** Divide the change by the original, 800.

A	Correct	$200 / 800 = 0.25$.
B	Confusion set	$800 / 1,000$ is the old value as a share of the new, not a change.
C	Inverted operation	Divides by the new value, 1,000.
D	Wrong stem number	Reports the raw change, 200, as a percent.

8. Answer A Concept 5 • transfer

After a 25% increase, a price is \$60. What was the original price?

Target word: solve for the original **Decides:** $\text{Original} \times 1.25 = \60 .

A	Correct	$\$60 / 1.25 = \48 .
B	Inverted operation	Takes 25% of the new price off; the base is the original.
C	Wrong stem number	Subtracts 25 dollars instead of 25 percent.
D	Confusion set	Applies another 25% increase.

9. Answer A Concept 6

In Room 14, students build a tile pattern: stage 1 uses 5 tiles, stage 2 uses 8, and stage 3 uses 11. If the pattern continues, how many tiles are in stage 20?

Target word: 20th term **Decides:** Adds 3 each stage: tiles = $3n + 2$.

A	Correct	$3(20) + 2 = 62$.
B	Dropped element	Drops the constant +2.
C	Camouflage	Uses the room number 14 as n.
D	Adjacent category	Treats the pattern as proportional (5×20).

10. Answer C Concept 7

A math coach observing a seventh-grade class that uses tablets reviews four sets of ordered pairs a student wrote. Which set is a function?

Target word: identify the function **Decides:** No input may map to two outputs; repeated outputs are fine.

A	Inverted operation	Input 2 has two outputs, the property that breaks a function.
B	Confusion set	Looks like a familiar curve, but input 1 has two outputs.
C	Correct	Each input has one output; the repeated output is allowed.
D	Absolute language	Outputs may repeat; only inputs are restricted.

11. Answer C Concept 7 · transfer

For $f(x) = 2x - 3$, $f(a) = 11$. What is a?

Target word: solve for the input **Decides:** $2a - 3 = 11$.

A	Inverted operation	Computes $f(11)$ instead of solving for the input.
B	Confusion set	Subtracts 3 instead of adding it back: $(11 - 3) / 2$.
C	Correct	$2a = 14$, $a = 7$.
D	Wrong stem number	The output, not the input.

12. Answer C Concept 8

During a lesson recorded for the teacher's portfolio, a student in a class of 24 says 0.25 is greater than 0.4 because 25 is greater than 4. What is the best response?

Target word: compare decimals **Decides:** $0.4 = 0.40$, which is 40 hundredths.

A	Adjacent category	Being in the same range does not make values equal.
B	Absolute language	Digit count does not decide size.
C	Correct	Place value, not digit count, decides.
D	Confusion set	Right answer, wrong reason; it fails for 0.5 and 0.52.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	4	
Wrong stem number	A figure copied from the prompt that is not the answer.	3	
Adjacent category	The right idea in the wrong bucket.	4	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	10	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	10	
Camouflage	Answers a stem fact that was never the question.	1	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Order of operations	Grouping first, then exponents, then multiplication and division left to right, then addition and subtraction left to right.	Always multiplying before dividing, or working strictly left to right.	A mixed expression with no parentheses.
2	Subtracting integers	Subtracting a negative is adding its opposite: $a - (-b) = a + b$; $\text{change} = \text{final} - \text{initial}$.	Adding the signed values, or dropping the sign.	A rise or gap measured from a value below zero.
3	Dividing fractions	Dividing by a fraction means multiplying by the reciprocal of the divisor.	Flipping the first fraction, or multiplying straight across.	"How many ... fit in ..." with fractional amounts.
4	Decimal-percent conversion	Percent means per hundred: move the decimal point two places right to go from decimal to percent.	Moving the point the wrong way or tacking on a % sign.	A decimal rate that must be stated as a percent.
5	Percent change uses the original as the base	Percent change = $(\text{new} - \text{original}) / \text{original}$; to undo a p% increase, divide by $(1 + p)$.	Dividing by the new value, or subtracting p% of the new value.	Two values in time order, or a value "after" an increase.
6	Linear patterns	A pattern that adds the same amount each stage has nth term = $(\text{constant difference}) \cdot n + (\text{first term} - \text{difference})$.	Treating a growing pattern as proportional (multiplying the first term).	Stage 1, 2, 3 counts with an equal jump.
7	What makes a function	A function assigns each input exactly one output; outputs may repeat.	Thinking outputs must all be different.	Sets of ordered pairs or a mapping to classify.
8	Decimal place value	Compare decimals by place value (align or add zeros), not by how many digits they have.	"Longer decimal is larger" or "shorter decimal is larger."	Two decimals with different numbers of digits.

Course scope (public description): Number Sense and Functions is a performance-based assessment that evaluates a student's portfolio of work. This portfolio includes the student's responses to various prompts and an original lesson plan for each of the mathematics modules such as number sense, patterns and functions, integers and order of operations, fractions, decimals, and percentages.